Fake News Detection using Social Context and Content Features

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Abstract

This project is designed to create a fake news detection system using the FakeNewsNet
(https://github.com/KaiDMML/FakeNewsNet) dataset that integrates both news content and social context features. Our group has the goal of building a classification function that can detect fake news articles,often said using combining textual analysis with user and propagation network information. We will untilize some machine learning models and technique such as logistic regression, random forests, and deep learning methods, and evaluate them on some benchmarks, such as including accuracy, precision, recall, and F1 score. The project also will have dataset preprocessing, feature engineering, model development, and evaluation at the end. The final result of our built system logistics will demonstrate the feasibility of detecting fake news by integrating both content-based and context-based signals.

ACM Reference Format:
Fred Hou, Yixiang Zhang, Xiaoxi Chen, and Xiaoxiong Lei. 2026. Fake News Detection using Social Context and Content Features. In . ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

The fast increase of the quantity of inaccurate information reveals a quite significant challenge to online websites and society at a big scale. Fake news detection is a very complicated task that requires our analysis not only on the textual content of the news' articles but also on the social interactions that surround the information. Unlike simpler classification tasks such as spam detection, fake news detection involves multi-modal data sources including images, text, etc. The FakeNewsNet repository provides a comprehensive dataset containing both the text content of news articles and the associated social context from some social media platforms such as Twitter, offer the opportunity to study the interactions between language and user behavior in identifying fake. By utilizing those learnt natural language processing (NLP) techniques with machine learning models techniques and graph-based features, our project is built to design a robust system that can effectively detect those news that is fake. This work will involve four group members, and we will work with the techiniques and capabilities that we gain in the class.

2 Description

a)Based on the current situation, our group has decided to develop a fake news detector. The which function is to provide a given news, conduct text retrieval, analyze the text, and judge the probability that the given news is fake news. Our group plans to adopt a machine learning approach and continuously train to develop a reliable model to achieve relatively accurate detection of false news.

b)This project will focus on the dataset from FakeNewsNet introduced by Shu et al. [1]. The dataset contains multiple components of news articles, including the label, statement, subject(s), speaker, speaker's job title, state information, party affiliation, and the speaker's credit history count (including the current statement). Such a large collection of information provides sufficient features to develop models for identifying the accuracy of input news. Furthermore, the dataset includes news from multiple sources such as news articles, tweets, and spatiotemporal information. The variety allows models to analyze information across different contexts and backgrounds, thereby improving the reliability of results.

c)After successfully training and obtaining a reliable model through the database, our group plans to use fresh data from multiple sources, including the most cutting-edge news from various groups on Tiktok, Facebook and X, to test the model and summarize and evaluate the results one week later. Rapidly updated social networking platforms are often the places where fake news is fabricated and spreads quickly to create buzz. Due to the complexity of information about rumors and fake news and the rapid development of AI technology, fake news has become even more difficult to distinguish between true and false, and the work of refuting them is extremely challenging. One week is the time point when most fake news is successfully refuted. Our group will select relatively easy-to-distinguish fake news and real news to test the model.

(d)Timeline and Workload Distribution per Teammate

We assign a distinct set of tasks to each of the four members to make sure the work is well balanced and completed in time. Although only one team member is assigned to each phase, other team members will provide support and learn together while this member leads the phase tasks.

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ACM ISBN 978-1-4503-XXXX-X/2018/06
https://doi.org/10.1145/nnnnnnn.nnnnnnn

2.0.1 Week 1-2 Sep 15 – Sep 26 Dataset Pre-processing & Setup Xiaoxi Chen. Xiaoxi will download the FakeNewsNet dataset, validate the file integrity, and unify a standard schema for the articles, users, and propagation traces. She will clean and normalize the text content. For instance, this will include language detection, lower-casing, URL and handle expansion, punctuation and emoji handling, stop-word policy, and lemmatization. Moreover, she will de-duplicate items and also cross-validate the labels against source metadata. In addition, she will also conduct exploratory analysis to profile class balance, topic distribution, source bias, and feature correlations, and she will provide a reproducible preprocessing script.

2.0.2 Week 3-5 Sep 29 – Oct 17 Feature Engineering Fred Hou . Fred will extract textual features using Term Frequency–Inverse Document Frequency (TF -IDF), which ranks the importance of a word to an article based on how frequently it appears there compared to how frequently it appears across all articles. He will also construct word embeddings which can be used to encode semantic similarity between words. To capture social context, he will design features including a user credibility score, as well as the number of shares and retweets, and the information cascades through which information propagates through the network.

2.0.3 Week 6-8 Oct 20 – Nov 7 Model Development Yixiang Zhang . Yixiang will develop traditional machine learning models such as logistic regression, which outputs the probability that an article is fake by fitting a linear function to the data, and random forest, which builds multiple decision trees to model more complex structures in the data. He will also develop deep learning models, such as neural networks that are able to work with both text and social context data. Yixiang will also tune the hyperparameters of these models, which involves settings like the learning rate, the depth of the decision trees, and the size of the neural networks.

2.0.4 Week 9-10 Nov 10– Nov 21 Evaluation & Testing with Real Data Xiaoxiong Lei . Xiaoxiong will develop the testing and evaluation pipeline. He will run the trained models on newly collected data from TikTok, Facebook, and X, taking care to sample responsibly and preprocess consistently. He will calculate accuracy, precision, recall, F1 score, area under the ROC curve, and report confidence intervals. He will perform error analysis to find common failure cases, e.g., topic drift and coordinated reposting, and verify robustness via ablation studies and stress tests on noisy inputs. He will document findings and suggest concrete fixes to improve the next training cycle.

2.0.5 Week 11 Dec 1 – Dec 5 Integration & Final Report All Members. We will write the final project report and prepare the presentation slides. Each of us will mainly focus on our own phase during the process of this project. Then we will make a presentation video to introduce our Fake News Detector.

3 Evaluation

Until now, our system use 504 pieces of political news articles in the FakeNewsNet dataset to do some data analysis and model training. We separate real news and fake news based on the techniques of text retrieval and text classification we learned in the class. This dataset includes 239 fake news articles(47.4%) and 265 real news article(52.6%), we split them into training group(70%), development group(15%), and test group(15%), we use stratified sampling to keep the groups balanced.

3.1 Baseline Model: TF-IDF + Logistic Regression

Our baseline model uses a classical machine learning approach that can actually combine both Term Frequency-Inverse Document Frequency (TF-IDF) vectorization with Logistic Regression. The feature set of our using consists of 5,507 dimensions: 5,000 TF-IDF features from article body text, 500 TF-IDF features from titles, and 7 stylometric features (text length, word count, punctuation ratio, uppercase ratio, etc.). This approach actually directly uses the Vector Space Model and term weighting concepts from the class we learned in the semester.

The most discriminative features, we said, detected by the baseline model reveal that is distinct patterns between those fake and real news, as stated in Figure 1. Fake news articles are found to tend to be shorter, use more sensational language, contain political figures' names prominently ("trump", "clinton", "obama"), and shows higher punctuation and uppercase ratios. In the contrasts, real news demonstrates professional journalistic language ("according", "statement", "official"), longer and more detailed content, and lower stylistic intensity.

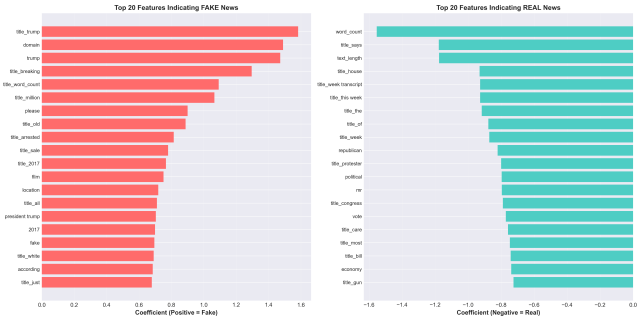


Figure 1: Feature Importance - Top predictive features for fake vs. real news in the baseline model. Positive coefficients indicate fake news markers, while negative coefficients indicate real news markers.

On the held-out test set of 76 articles of ours done, the baseline model actually achieved an accuracy of 86.84%, significantly exceeding our initial target of 80% and outperforming the majority-class baseline (52.6%) by 34.2 percentage points.Besiseds, The model demonstrated balanced performance across both classes, that with macro-F1 score of 0.868, correctly identifying 85.0% of real news and 88.9% of fake news.

3.2 Advanced Model: DistilBERT Transformer

We finally fine-tuned a DistilBERT transformer model (distilbert-base-uncased) on the same dataset splits. DistilBERT is a model we used that is a distilled version of BERT with 66 million parameters.

It actually definitely uses the so called attention mechanisms to capture contextual semantic relationships. we noticed that traditional bag-of-words models cannot represent. The model was trained, as it with for 2 epochs and by using a learning rate of 2e-5 and batch size of 4, and a maximum sequence length of 256 tokens.

The DistilBERT model actually achieved substantially higher performance on the test set. We indeed observe a 92.11% accuracy with macro-F1 of 0.921, and also representing a 5.27 percentage point improvement over the baseline. More importantly, the transformer model indeed reduced classification errors from 10 to 6 articles (40% error reduction). This results are demonstrating superior generalization capability.

3.3 Comparative Analysis

Figure 2 presents side-by-side confusion matrices for both models on the data that ours. The baseline model actually correctly classified 34 real and 32 fake articles. And the DistilBERT model can improved to 36 real and 34 fake articles correctly identified. And we find out that DistilBERT achieved 90.0% recall on real news (up from 85.0%) and 94.4% recall on fake news (up from 88.9%).All these indicating more reliable detection across all categories of it.

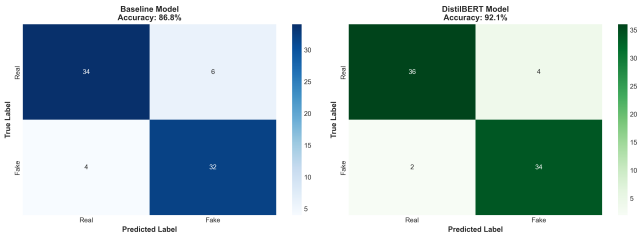


Figure 2: Confusion Matrix Comparison - Side-by-side view of baseline (left) and DistilBERT (right) classification performance. DistilBERT demonstrates fewer false positives and false negatives.

The ROC curves in Figure 3 shows that both models' excellent discrimination ability. We also noticed that with the baseline model of ours achieving an AUC of 0.926 and DistilBERT reaching 0.972. Therefore, we can say that this 0.046 improvement indicates that DistilBERT indeed assigns higher probabilities to those that true positives. And as well, lower probabilities to true negatives more consistently. Similarly, as well as the Precision-Recall curves (Figure 4) show DistilBERT's Average Precision of 0.966 compared to the baseline's 0.898. This demonstrating superior precision-recall trade-off across all decision thresholds.

Figure 5 provides a comprehensive overview of all evaluation metrics. DistilBERT outperforms the baseline across every dimension: accuracy (+5.3%), macro-F1 (+5.3%), ROC-AUC (+4.7%), and PR-AUC (+6.8%). These consistent improvements across diverse metrics validate the effectiveness of transformer-based deep learning for fake news detection.

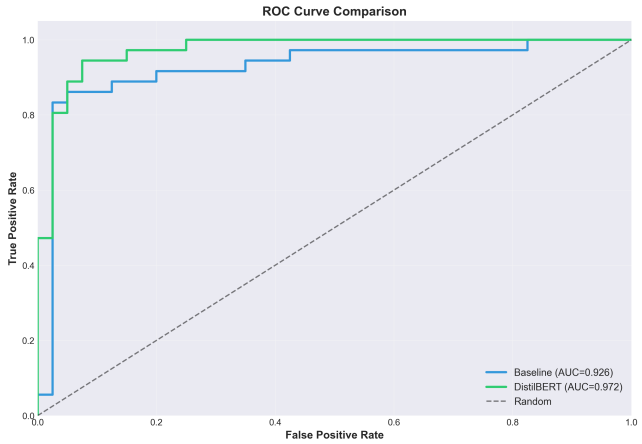


Figure 3: ROC Curve Comparison - Both models demonstrate excellent discrimination ability (AUC > 0.92), with DistilBERT achieving near-perfect separation (AUC = 0.972).

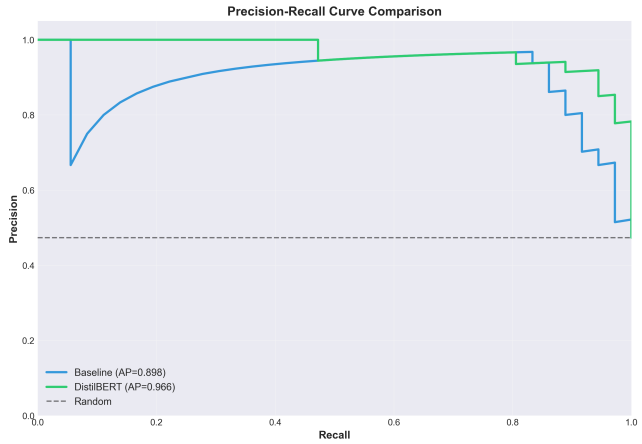


Figure 4: Precision-Recall Curve Comparison - DistilBERT maintains higher precision across all recall levels, with Average Precision improving from 0.898 to 0.966.

3.4 Error Analysis

Detailed error analysis reveals some reulst. That 84.2% of test articles (64/76) received identical predictions from both models. Therefore we can find that result is indicating strong agreement on clear-cut cases. DistilBERT successfully corrected 6 baseline errors. This includes political transcripts and opinion pieces that the baseline model incorrectly flagged as fake due to their non-traditional journalistic format. However, DistilBERT introduced 2 new error on artilecs with complex political advertising language. Therefore, we can say four articles remained misclassified by both models. They are all indeed representing genuinely ambiguous cases. We have examples, such as short political announcements and investigative journalism, that mimics fake news stylistic patterns.

Figure 6 visualizes the error reduction achieved by DistilBERT. The net improvement of 4 correctly classified articles (from 66

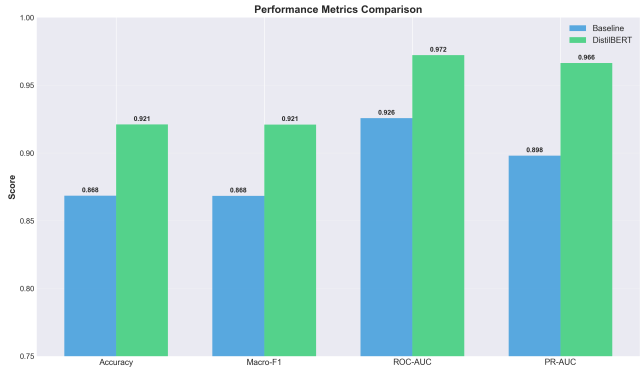


Figure 5: Performance Metrics Comparison - Bar chart showing DistilBERT's superiority across all evaluation criteria. All metrics exceed 0.92, indicating excellent model performance.

to 70 correct predictions) represents a 40% reduction in errors, demonstrating the practical value of employing transformer models for this task.

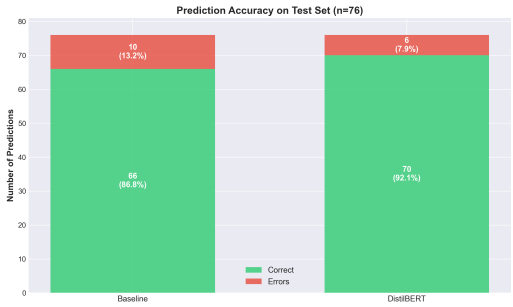


Figure 6: Error Reduction Analysis - DistilBERT achieves 92.1% accuracy (70/76 correct) compared to baseline's 86.8% (66/76 correct), reducing classification errors by 40%.

4 Discussion

In this experiment, we conclude that both methodologies are effective with tradeoffs that are evident in their results. The TF-IDF + Logistic Regression baseline achieved an accuracy of 86.84%. This result showcases that lexicon and stylometry are good indicators to perform the detection task of political fake news in social media. We can make this statement by the brief analysis of the feature importance in this section as, on the one hand, fake news have consistent signals which are linguistic artifacts: short in length, containing high punctuation ratio, and most political divisive terms. On the other hand, real news is more consistent with journalistic patterns. This result is also aligned with state of the art works as mis-information has a big presence of stylometric elements (Shu et al., 2019).

On the other hand, DistilBERT model was able to improve this baseline, by achieving a higher accuracy and improving the macro-F1, ROC-AUC, and PR-AUC by a considerable margin. This can be seen by the advantage of contextualized word representations which capture semantic and syntactic relationships which a simple bag-of-word cannot provide. We noticed that DistilBERT was able to correct most of the errors from the baseline model. We specifically observed that misclassified news from the baseline model are composed of article which are opinion in nature and in a format similar to an interview. It is much harder to discern between this kind of articles and real mis-information articles using stylometric analysis as they also use hyperbolic or divisive language in their content. The errors in total were also reduced by 40% in percentage, which is a substantive improvement.

Despite achieving these results, we found that the models were not perfect. Both models have difficulties discerning ambiguous one-liner political statements and statements. We also observed that the models had problems with political advertisements and articles which mimic an objective reportage format with a very high ratio of a persuasive framing, this combined signal of information and opinion makes it difficult to classify it into either one of the two classes. We also found that mis-information which have a high credibility user tend to fool the models, this may be in a false signal as high credibility users retweet or create such kind of tweets which showcase that these mis-information with high propagation cannot be considered as fake news using content alone. DistilBERT also made some minor errors where the style of a rhetorical figure in the test set was not represented in the training set.

With these considerations in mind, we further recommend the following actions to build on these results: Adding more social context features as features in this model. We can use social propagation, user level features such as user network information etc. We can also train this models in an online fashion with datasets like GossipCop as this will allow the model to learn more about misinformation and reduce the risk of topic drift and domain shift. We also find that the experiments and the model needs to be trained on more domains other than political misinformation as we found in our error analysis. This should allow the model to be more robust and cross-domain in this detection task. The addition of graph based models or use of credibility based features in an ensemble method along with the transformer representations might allow the model to be more robust over platforms.

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